

DOTE 6635: Artificial Intelligence for Business Research

Final Words

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Session	Date	Topic	Key Words
1	1.06	AI/ML in a Nutshell	Course Intro, Prediction in Biz Research, ML Model Evaluations
2	1.13	Intro to DL	DL Intro, Neural Nets, DL Computations
3	1.20	RL (I)	Tabular MDP, Bellman Equation
4	1.27	RL (II)	Model-free Prediction & Control, Monte Carlo, Temporal Difference
5	2.03	RL (III)	Value-based DRL, DQN, AlphaGo
6	2.10	RL (IV)	Policy-based DRL, Policy Gradient, PPO, A3C
7	2.24	RL (V)	Off-Policy Evaluation, Off-Policy Learning
8	3.3	RL (VI)	RLHF, RL with Verifiable Reward, LLM Reasoning, GRPO
9	3.10	Agentic AI (I)	Building LLM Agents, ReAct, ToT, MCP
10	3.17	Agentic AI (II)	Agent Training, Tool Use, Planning, Coding Agent
11	3.24	Agentic AI (III)	Agent Evaluation, Benchmarks
12	3.31	Agentic AI (IV)	Multi-Agent Systems, Agentic Simulations
13	4.14	Agentic AI (V)	Business Applications of Agentic AI
13+	2027	Season 4	More to come...

What Happened in the Past 3 Months?

A lot of **Reinforcement Learning**.

Decent amount of **Agentic AI**.

Decent amount of their **applications in biz/econ research**.

Great emphasis on the **frontier of AI technology**.

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Application of AI/ML in Business Research

Reinforcement learning **scales the reasoning capability of LLM** and, therefore, empowers modern language agents.

AI agents have driven a substantial **paradigm shift** in research.

Setting & Problems

ML as Subjects

Data Generation

ML as Data Source

OR for LLM Inference

Agentic AI could **automate any knowledge work**.

Hypothesis Testing

ML for Causal Inference

Mechanisms

Prediction/ Optimization Models

ML for Predictive Decision-Making Optimization

Decisions

Structural Models

ML for Structural Estimation

Counterfactuals

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Our Goal

1. Have a basic understanding of the **fundamental concepts/methods** in machine learning (ML) and artificial intelligence (AI) that are used (or potentially useful) in business research.
2. Understand how business researchers have utilized ML/AI and what **managerial questions have been addressed by ML/AI** in the recent decade.
3. Nurture a taste of what the **state-of-the-art AI/ML technologies** can do in the ML/AI community and, potentially, in your own research field.



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Course Takeaways

- The **necessary knowledge** of AI/ML that could help you:
 - Keep up with the **literature development** in the relevant domains in both CS and business;
 - Develop the **necessary sense** to do **rigorous business research** using the relevant methods;
 - Identify **important and interesting questions** in your own field where AI technologies are useful;
 - Invent new applied methods** (most likely without any theoretical guarantee) in your own research.
 - Build **agentic pipelines and workflows** to automate your own research and achieve 10x productivity.

Impact of a **CS** Paper = Problem Importance * **Technical** Novelty * **Performance** Improvement

Impact of a **Business** Paper = Problem Importance * **Identification** Rigor * **Insight** Novelty

- Academic research is a kind of **craft**: You can only learn by **doing it on your own**.
 - So, take your **final projects (or some other AI-related research)** seriously!

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When Will Things Go Wrong?

- Most AI applications are only useful if actionable insights can be derived:

$$\frac{d\pi(X_0, Y)}{dX_0} = \frac{\partial \pi}{\partial X_0} \underbrace{(Y)}_{\text{prediction}} + \frac{\partial \pi}{\partial Y} \underbrace{\frac{\partial Y}{\partial X_0}}_{\text{causation}}$$

Your prediction of Y is not accurate.

Your causal identification is not clean.

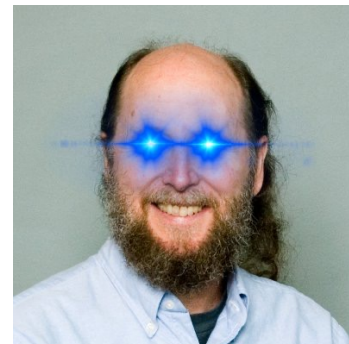
- You should be able to judge whether you should seek for **accurate prediction** and/or **clean identification**.
- Empirical model: $Y = a + b \cdot D + g(X) + \epsilon$
 - Key parameter of interest: b
 - If D is predicted by a ML model, the prediction error is likely to be correlated with ϵ , giving rise to the bias to estimate b .
 - If $g(X)$ is predicted by a ML model, you may leverage the DML framework to remove the regularization bias and overfitting bias.

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The Bitter Lesson

- References: <http://www.incompleteideas.net/IncIdeas/BitterLesson.html>
<https://www.youtube.com/watch?v=vbVfAqPI8ng>
- The biggest lesson that can be read from 70 years of AI research is that **general methods that leverage computation** are ultimately the most effective, and by a large margin.
- Leveraging domain knowledge (short-term & specific) vs. Leveraging computation (long-term & general).
- Bitter lesson: Leveraging domain knowledge is **self-satisfying** and **intellectually inspiring**, but plateaus in the long-run or even inhibits further progress.
- Are you ready to control the machine intelligence to create great knowledge?**



Prof. Richard Sutton

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What to Expect?

- Summer 2026 Reading Group on AI x empirics.
 - <https://github.com/rphilipzhang/Reading-Group-2026>
- What to expect in Spring 2027:
 - Agentic AI
 - Multimodal LLM
- Stay tuned and hope to see you all again!

Overview

This reading group has two parallel tracks, each occupying one hour per session:

Time	Track	Focus
9:30 - 10:30	Applied Empirics	Core causal inference and econometric methods applied in business/economics research
10:30 - 11:30	AI for Research	Practical AI techniques that power modern research workflows

Prerequisites & Assumptions

- Applied Empirics:** Participants are expected to have a solid foundation in econometrics and causal inference (e.g., OLS, potential outcomes, basic DID, IV, RDD concepts). The reading group focuses on **applying** these methods **correctly** in empirical research settings, navigating practical challenges such as assumption validation, robustness checks, and replication, rather than re-teaching the underlying theory.
- AI for Research:** Participants are assumed to be familiar with the basics of AI and LLMs (e.g., what a large language model is, how to use ChatGPT-style interfaces for simple tasks). The reading group focuses on **modern agentic AI**, multi-agent workflows, autonomous research systems, AI coding agents, and integration of AI into end-to-end research pipelines, which goes well beyond basic LLM usage. We will build AI systems together to empower our research.

Track Descriptions

Applied Empirics follows the structure of [Goldsmith-Pinkham's Applied Methods course](#) and [Sant'Anna's DID resources](#), covering canonical identification strategies (DID, IV, RDD) and modern ML-based causal inference methods, with emphasis on assumptions, simulations, and hands-on replication.

AI for Research draws on recent developments in agentic AI systems, AI coding assistants, and autonomous research pipelines, aiming to equip participants with practical skills to integrate modern AI tools into their own research workflows.

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Keep in Touch

- Stay in contact and keep me posted of your academic and career successes.
- Feel free to send me an email/WeChat message. I am always happy to discuss topics related to AI technology, business research, and business.
 - We may work on something interesting together ☺
- Let me know if you need a job referral from me to comment on your academic/career potential.

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The Thinking Game

Reference: <https://www.bilibili.com/video/BV1bbU8BREog/?t=1468s>



Sir Demis Hassabis

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Finally

Thank You & All the Best!

谢谢!
祝前程似锦!

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Hope to see you all again!

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